

Vaibhav Satish

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Education

University of California, Irvine: Donald Bren School of Information and Computer Science

Irvine, CA

Bachelor of Science(B.S) Computer Science

Aug 2024 - Mar 2027

- GPA: 3.873
- Achievements: Dean's Honors List (6/6 consecutive academic quarters)

Work Experience

Software Engineering / Machine Learning Intern

Irvine, CA

Stealth AI Startup

Jun 2026 - Present

- Integrated an **OpenAPI API documentation system** utilized by the entire development team, including the **CTO and senior engineers**, by dynamically generating endpoint details from **JSDoc comments**.
- Improved measurement pipeline to process **13** outputs from a **Multilayer Perceptron model**, geometrically calculate **8** length-based measurements, and derive **4** additional circumference measurements using an **anthropometric prior fusing system**.
- Seeded **NeonDB** and **GCP Cloud SQL** databases across **10+** backend endpoints and shipped an optional **brand-filtering** feature enabling brand-specific queries platform-wide.

Software Developer Research Assistant

Irvine, CA

UC Irvine Charlie Dunlop School of Biological Sciences

Mar 2026 - Present

- **Scaled the MedTech@UCI cognitive project into a formal lab partnership**, integrating our cognitive game into the MindCycle app for clinical trials.
- Collaborated with the **UCI SLEEP Lab** under **Dr. Katherine Simon**, tracking **4+** core behavioral metrics via **backend logging**.

Software Development Co-Lead

Irvine, CA

MedTech@UCI

Sep 2024 - Present

- Co-lead a team of **5** engineers using **Unity** and **C#** to develop a pediatric cognitive analysis app with image-recognition and memory tools, enabling earlier detection of cognitive delays.
- Designed interactive cognitive games using **React** and **JavaScript**, delivering therapeutic interfaces for dementia and Alzheimer's patients.
- Engineered a **React metrics-tracking engine** to capture **15+** core cognitive metrics, providing clinicians with actionable analytical insights.

Software Development Intern & Instructor

Seattle, WA

Tech Academy of Minnesota

Jun 2025 - Aug 2025

- Delivered **full-stack** curriculum to **18** students, guiding web development using **Python**, **HTML**, **CSS**, and **JavaScript**.
- Formulated structured **debugging** frameworks and **12** lab exercises, accelerating core engineering and algorithmic problem-solving.
- Translated complex **software engineering paradigms** into digestible concepts, providing students with a strong foundation of programming fundamentals.

Artificial Intelligence Researcher

Remote

Stanford University (via Lumiere Research Program), Georgia Tech & Cornell University (via Inspirit AI Research Program)

May 2023 - Nov 2023

- **Developed, trained, and tuned multi-modal predictive machine learning** models—including **Softmax Regression** and **Deep Neural Networks**—using **PyTorch**, **Scikit-Learn**, **NumPy**, and **Pandas** on a **270,000-row** NFL injury dataset, improving predictive accuracy by **14%** over baseline models.
- Trained **3+ TensorFlow** and **Scikit-Learn** computer-vision models to detect distracted driving, achieving **80-85%** accuracy, and optimized inference pipelines for real-time edge deployment.

Projects

ZOTNest - UC Irvine's 2026 IrvineHacks Hackathon

Feb 2026 - Present

React, Python, FastAPI, Supabase, Gemini Embeddings, RAG, Leaflet

- Built a full-stack property search app for 30,000+ UCI students, combining **multi-criteria filtering** with a **Retrieval-Augmented Generation (RAG)** pipeline for semantic search.
- Embedded structured listing data and scraped Google reviews per property via **Gemini embeddings**, storing vectors in a **Supabase pgvector** store for similarity search.
- Combined **LLM-based preference extraction** (budget, bedrooms, distance, shuttle) with query embeddings into a single hybrid relevance score.
- Integrated Leaflet for real-time property mapping, and scraped 100+ Google reviews to power per-property "vibe" search.

Wordle

Jun 2026 - Present

JavaScript, Python (FastAPI), CSS, React, Next.js

- Deployed a **Wordle** clone with a **FastAPI** backend handling word validation, answer selection, and game-state logic via a **REST API**.
- Built a responsive **React** frontend with real-time keyboard-state tracking, color-coded feedback, and **WordNet**-based word-definition lookups.

Driver Delineation - UC Irvine's 2025 Datathon

Apr 2025

Python, PyTorch, TensorFlow, Scikit-Learn, NumPy

- Trained supervised **ML models** using **PyTorch** and **Scikit-Learn** to predict driver likelihood from high-dimensional datasets.
- Executed **feature engineering** on **40+** impact variables to minimize model overfitting and boost generalizability.
- Designed a multi-layer **Deep Neural Network (DNN)** in **TensorFlow**, tuning hyper-parameters to hit a **90%** validation accuracy.

UCI Themed Geoguessr

Dec 2024 - Present

React, Node.js, TypeScript, C++, Crow API, Google Cloud Platform, Vercel

- Led a **4-person** engineering team from **system architecture ideation** through deployment, managing structured Git workflows.
- Constructed a custom mathematical scoring algorithm and integrated **Leaflet** for sub-second asset rendering and map integration.
- Architected an asynchronous **C++ backend** using **Crow API** and **Google Cloud Platform** to minimize gameplay latency and handle concurrent asset updates.

Technical Skills

Languages/Frameworks: Java, Python, C++, SQL, TypeScript, JavaScript, React, Node.js, Next.js, FastAPI, REST APIs, OpenAPI, C#

AI/ML: Machine Learning, Supervised/Unsupervised Learning, NLP, Computer Vision, PyTorch, TensorFlow, Scikit-Learn, Data Analysis

Databases/Cloud Infrastructure: MySQL, PostgreSQL, Google Cloud Platform

Other: HTML, CSS, Docker, Git, Unity, Agile, CI/CD, Data Structures, Algorithms, Software Engineering Paradigms